

Detecting primate vocalizations in long tropical-forest recordings

A VGG19 frequency-position CRNN with two-stage transfer learning, a hard-negative confuser class, and automatic false-positive cleanup

Reference clips

Cernic + Colobus
+ confuser class
+ Background
(4 classes)

Mel-spectrogram + augmentation

128 mel bins,
20–8 000 Hz; 7× aug;
HF-nuisance aug
for Colobus (V12)

VGG19 + CRNN classifier

frequency-position
head (CoordConv) →
per-band Conv1D →
BiLSTM → softmax

Sliding-window detection

2 s / 1 s windows,
group → threshold →
NMS → low-freq
gate (Colobus)

Three-filter auto-cleanup

Mahalanobis OOD
+ YAMNet check
+ temporal
isolation

Hard-negative mining

confirmed false
positives folded
into Background
for retraining

iterative hard-negative loop: detect → clean → fold into Background → retrain

Two-stage transfer learning · 98.12 % validation accuracy · near-zero primate cross-confusion · automatic, near-zero manual listening